Electrolysis Question Bank

Practice Questions

8 problems • 3 topics

Faraday's Law • Selective Discharge • Anodising

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Questions-only edition

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1. Faraday's Law: Deposition and Gas Evolution

Data for calculations. Take the Faraday constant as $F = 96\,500\,\mathrm{C\,mol^{-1}}$ and, where required, the molar volume of a gas at room conditions as $24.0\,\mathrm{dm^3\,mol^{-1}}$.

Question 1 — Mass of Copper Deposited

A current of 2.00 A is passed through $\mathrm{CuSO_4(aq)}$ using copper electrodes for 20.0 minutes.
Calculate the mass of copper deposited at the cathode.

$$\mathrm{Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)}$$
$$A_r(\mathrm{Cu}) = 63.5$$

Question 2 — Time Needed to Deposit Silver

A student wants to deposit 2.16 g of silver from $\mathrm{AgNO_3(aq)}$ using a current of 1.50 A.
Calculate the time required.

$$\mathrm{Ag^+(aq) + e^- \longrightarrow Ag(s)}$$
$$A_r(\mathrm{Ag}) = 108$$

Question 3 — Volume of Hydrogen Produced

A current of 3.00 A is passed through dilute $\mathrm{H_2SO_4(aq)}$ using platinum electrodes for 15.0 minutes.
Calculate the volume of hydrogen gas produced at the cathode.

$$\mathrm{2H^+(aq) + 2e^- \longrightarrow H_2(g)}$$

Question 4 — Unknown Elements

A current is passed through two cells connected in series.
The first cell contains $\mathrm{XSO_4(aq)}$, while the second cell contains $\mathrm{Y_2SO_4(aq)}$.
The relative atomic masses of X and Y are in the ratio $1 : 3$.
What is the ratio of the mass of X liberated to that of Y?

Question 5 — Two Cells Connected in Series

Two electrolytic cells are connected in series.
Cell 1 contains $CuSO_4(aq)$, and copper is deposited:

$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

Cell 2 contains $AgNO_3(aq)$, and silver is deposited:

$$Ag^+(aq) + e^- \longrightarrow Ag(s)$$

A current is passed until 0.635 g of copper is deposited in Cell 1.
Calculate the mass of silver deposited in Cell 2.

$$A_r(Cu) = 63.5, \qquad A_r(Ag) = 108.$$

2. Selective Discharge and Electrode Potentials

Question 6 — Order of Metal Plating

A solution at 25 °C contains 1.0 M Cd^{2+}, 1.0 M Ag^+, 1.0 M Au^{3+}, and 1.0 M Ni^{2+} in the cathode compartment of an electrolytic cell.
Predict the order in which the metals will plate out as the voltage is gradually increased.

Question 7 — Selective Plating under Non-standard Conditions

A solution at 25 °C contains 1.0 M Cu^{2+} and 1.0×10^{-4} M Ag^+.
Which metal will plate out first as the voltage is gradually increased when this solution is electrolysed?

Hint. Use the Nernst equation to calculate the reduction potential, E, for each half-reaction.

3. Anodising Aluminium

Question 8 — Anodising Aluminium

The process of anodising aluminium objects is one of the most common applications of electrolysis.

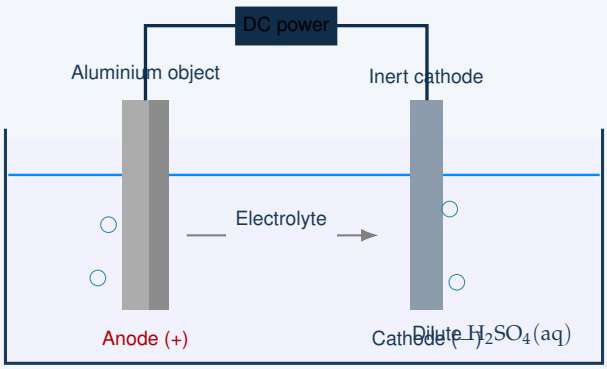


Figure 1: Schematic of anodising an aluminium object in dilute sulfuric acid.

The end result of this process is an increase in the thickness of the Al_2O_3 layer at the anode. However, consider the following standard electrode potential values:

$$Al^{3+}(aq) + 3e^- \longrightarrow Al(s) \qquad E = -1.66\,V$$
$$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l) \qquad E = +1.23\,V$$

It would seem that the E values favour the preferential oxidation of aluminium at the anode. Instead, the process takes place via the oxidation of water first.
Offer an explanation. Finally, state the two reactions that occur at the anode to eventually form the final Al_2O_3 product.

End of question bank